

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch: Mechanical Engg/Production Engg/Automobile Engg

Semester: VI

Subject Code & Name: IC Engines (BTMPE603A)

Max Marks: 60

Date: 23/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Compression ratio in diesel engine is	CO2	1
	a) 5-7 b) 7-10 c) 10-1-2 d) 14-20		
2	Carburetor is mainly employed in	CO1	1
	a) SI Engine b) C I Engine c) Gas Engines d) none of the above		
3	Knocking take place in CI Engine	CO4	1
	a) at the Start of combustion b) at the end of combustion c) during combustion d) none of the above		
4	For a four-cylinder vertical engine the commonly used firing order	CO4	1
	A) 1-2-3-4 b) 3-4-1-2 c) 1-3-4-2 d) 4-3-2-1		
5	Fuel injector is used for	CO3	1
	a) gas engine b) CI Engine c) SI Engine d) none of the above		
6	The most commonly used lubrication system in automobile is the	CO4	1
	a) Petrol system b) splash system c) Pressure system d) All of the above		
7	Supercharging is mainly done in	CO5	1
	a) Racing car b) Marine engine c) automotive diesel engine d) None of the above		

8	Indicated power is directly proportional to				CO4	1
	a) Torque	b) air consumption	c) cylinder peak pressure	d) none of the above		
9	Major constituents of natural gas				CO6	1
	a) ethane	B) Methane	c) propane	d) butane		
10	Blue smoke in diesel engine indicates				CO6	1
	a) NOX	b) HC	C) CO	d) Unburnt oil		
11	Purpose of fuel cell				CO6	1
	a) produce electric energy	b) propel the vehicle	c) provide fuel to the engine	d) all the above		
12	Main advantage of electric vehicle				CO6	1
	a) high power output	b) low running cost	c) provide more electrical energy	d) provide electricity		
Q. 2	Solve the following.					12
A)	Compare two stroke engine and four stroke engine.				CO1	6
B)	Explain valve timing diagram for high-speed SI Engines.				CO1	6
Q.3	Solve the following.					12
A)	What is detonation in SI engines with neat sketch.				CO4	6
B)	What are the combustion chambers used in CI Engines. Explain any two with neat sketch. https://www.batuonline.com				CO4	6
Q. 4	Solve Any Two of the following.					12
A)	Describe battery ignition system with neat sketch.				CO3	6
B)	What is supercharging and turbocharging?				CO4	6
C)	A single cylinder engine running at 1800 rpm develops a torque of 8 Nm. The indicated power of the engine is 1.8. kW. Find the loss due to friction power as the percentage of brake power.				CO5	6
Q.5	Solve Any Two of the following.					12
A)	Explain morse test for measurement of frictional power.				CO5	6

B)	Describe exhaust gas recirculation in engine emission controls.	CO6	6
C)	What are the different alternative fuels?	CO6	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the concept of fuel cell technology.	CO6	6
B)	What are the advantages and drawbacks of electric vehicle.	CO6	6
C)	Draw and explain layout of hybrid vehicle.	CO6	6
	*** End ***		